Exam in GRA 4160 (Spring 2024)

Predictive modelling with machine learning

30-hour take-home exam

29.05.2024 09:00 - 30.05.2024 15:00

Instructions

Attached to this exam are the following files:

1. `Exam_GRA4160_Spring2024.ipynb` (this file)
2. `diabetes.csv`
3. `Sales-Win-Loss.csv`

The assignments are divided into four sections:

1. Predict diabetes
2. Synthetic Data Generation and Ensemble Methods
3. Predicting Sales Performance Using a Neural Network
4. Reflection on Your Mini Project

Before starting, thoroughly read through all assignments. At the beginning of each added cell, explicitly state the exercise you are addressing. Ensure the cells are arranged logically and that all code cells execute error-free.

Ensure you document your code with comprehensive comments and explanations to keep it clean and readable. **Additionally, strive to make your non-code responses as concise and short as possible.** Long responses with irrelevant information can lead to a deduction of points. Preferably, put your non-code responses in markdown cells.

If you need further clarification on any assignments, please email Vegard H. Larsen at `vegard.h.larsen@bi.no`.

Remember to save and back up the notebook periodically.

Allowed Aids

- All course material and your notes.

- You can use the 3rd party Python libraries we used in class without any explanations (e.g., NumPy, Pandas, Scikit-learn, and Keras).
- Any other 3rd party Python libraries, but you must explain what they do and why you choose to use them.
- Resources on the Internet, but if you include code pieces you find on-line you must list the URLs of these resources in the corresponding resources' cell at the end of each section.
- You are NOT allowed to receive help from other people in solving the assignments.

Use of generative AI

You are allowed to use generative AI tools to generate code snippets, but you must explain the code snippets in your own words. Furthermore, you can use generative AI as an aid for all exercises, but you must explain why you chose to use generative AI tools and how you used them. If you use generative AI tools, you must list the tools you used in the corresponding resources' cell at the end of each section.

Submission

Submit your completed notebook (the `.ipynb` file) via WISEflow before the deadline. Be aware that the deadline is strict, so do not wait until the very last minute to submit.

Fill out your student ID at the designated area below.

Suggested grade boundaries (out of 100 points):

0 - 40 points: F

41 - 60 points: E

61 - 70 points: D

71 - 80 points: C

81 - 90 points: B

91 - 100 points: A

Assignment 1: Predict diabetes

This assignment is based on the Pima Indians Diabetes Database. The objective of the dataset is to diagnostically predict whether a patient has diabetes, based on certain diagnostic measurements.

The dataset is stored in the file `diabetes.csv` attached to the exam. The data can also be downloaded from **Kaggle**.

This dataset is originally from the National Institute of Diabetes and Digestive and Kidney Diseases. Several constraints were placed on the selection of these instances from a larger database. In particular, all patients here are females at least 21 years old of Pima Indian heritage.

The dataset consist of several medical predictor variables and one target variable. Predictor variables include the number of pregnancies the patient has had, their BMI, insulin level, age, and so on.

Exercise 1.1: Load the dataset from the file `diabetes.csv` and display the first five rows of the dataset. Comment on the structure and the completeness of the dataset. Check for missing values in the dataset. If there are any missing values, find a way to handle them. Briefly explain your approach.

Exercise 1.2: Explore the dataset by calculating the mean, median, standard deviation, and range of the numerical variables. Calculate the frequency of the categorical variables. Which variables do you think are most likely to be important for predicting diabetes?

Exercise 1.3: Choose two machine learning models for predicting diabetes. Discuss the reasons for choosing these models, including their strengths and weaknesses in the context of this specific problem.

Exercise 1.4: Split the data into training and testing sets. Train your models (the two models you selected in 1.3) on the training data and validate them using the testing data. Apply cross-validation and explain its importance in model validation.

Exercise 1.5: Evaluate the performance of each model using appropriate metrics. Present the results in a comparative format and discuss which model performs best and why.

Exercise 1.6: Interpret the results of your best-performing model. Discuss which features are most influential in predicting diabetes and why. Reflect on any limitations or biases in your model and how they could impact the results.

Guidelines

This assignment counts for 25% of the total grade.

Guidelines 1.1 (3 points)

- Check that the dataset is correctly loaded from `diabetes.csv` and the first five rows are displayed. Award points for correct use of the data loading function and accurate display.
- In this dataset missing values appear as zeros. Award points for identifying and handling missing values correctly.
- Evaluate the method used for handling missing values, including justification of the chosen method. Points should be awarded based on the appropriateness and implementation of the method (e.g., imputation, removal of missing data).

Guidelines 1.2 (4 points)

- Confirm calculations of mean, median, standard deviation, and range for numerical variables are correct. Evaluate the clarity and accuracy of the frequency analysis for categorical variables.
- Look for a thoughtful analysis or hypothesis about which variables might be important for predicting diabetes based on the statistics provided.
- Award points for clear presentation of statistics and insightful interpretation that supports subsequent analysis steps.

Guidelines 1.3 (4 points)

- Evaluate the choice of two machine learning models based on their suitability for the dataset and problem. The reasons should include the models' advantages and disadvantages.
- Look for a well-reasoned justification, including how each model's strengths and weaknesses align with the characteristics of this specific dataset.
- Award points for demonstrating understanding of the chosen models and their mechanisms.

Guidelines 1.4 (4 points)

- Check for correct methodology in splitting the dataset into training and testing sets, including the use of appropriate functions and parameters.
- Evaluate the implementation of cross-validation, ensuring it's applied correctly and effectively to both models. Assess the explanation of its importance in model validation.
- Check the validation process and clarity in the explanation of the methods used.

Guidelines 1.5 (5 points)

- Check that appropriate metrics (e.g., accuracy, precision, recall, F1-score) are used to evaluate each model.
- Points should be given for a comparative presentation of results that is clear and informative.
- Look for a thoughtful discussion on which model performs best and the reasons why, incorporating insights from the performance metrics.

Guidelines 1.6 (5 points)

- Evaluate how well the results and influence of different features in predicting diabetes are interpreted and explained.
- Give points for reflections on any limitations or biases in the model or data that could impact results.
- Grade the depth and critical analysis in discussing the model's performance and potential real-world implications.

Assignment 2: Synthetic Data Generation and Ensemble Methods

In this assignment, you will generate a synthetic dataset designed to simulate a real-world e-commerce scenario. You will then apply various tree-based machine learning models to this dataset to understand their characteristics and performance differences.

Part 1: Synthetic Data Generation

Use Python libraries such as `numpy`, pandas, and `scikit-learn` to generate a dataset of at least 10,000 instances. By instance, we mean a row in the dataset representing a customer and their behavior on an e-commerce website. Each instance should have an outcome variable indicating whether the customer made a purchase. Consider the following guidelines when creating the dataset:

- Create features typical for an e-commerce setting:
  - Age: Numeric, range between 18-70.
  - Gender: Categorical, Male or Female.
  - Income: Numeric, based on a normal distribution scaled to the economy of the scenario.
  - Number of website visits in the past month: Integer, consider a realistic range (e.g., 0-30).
  - Average spending per visit: Numeric, should correlate positively with income.
  - Time spent on website per visit: Numeric, in minutes.
  - Number of purchases in the last year: Integer, likely correlated with the number of visits.
- Define the target variable:
  - Purchased: Binary (Yes/No), could be influenced by the above features with added randomness for unpredictability.

You should make some assumptions about the willingness to purchase a product based on features defined above. Ensure your data includes realistic distributions when creating relationships between the features (e.g., higher income might correlate with higher spending per visit). Provide a brief explanation of your data generation strategy, especially how you ensure the realism of the data and the relationships between features.

Importantly, create some relationships between features that are not entirely linear. This will allow us to test the models' ability (in Part 2) to capture non-linear patterns in the data. Visualize the relationships between features using scatter plots or other appropriate visualizations.

Part 2: Tree-Based Models

In this part, you will train three different tree-based models on the synthetic dataset you created in Part 1.

- Train three different models of the following types:
  - Decision Tree
  - Random Forest
  - One additional ensemble model of your choice (e.g., XGBoost, or Bagging Classifier)
- Configure basic parameters for each model and document your choices.
- Discuss why each model might be appropriate for the dataset.
- Evaluate each model on the test set using your choice of metrics (e.g., accuracy,

precision, recall, F1 score, ROC-AUC). Briefly explain the reasons for choosing these metrics.
- Comment on the different models' ability to capture the non-linear relationships you introduced in the data.

Guidelines

This assignment counts for 25% of the total grade.

Guidelines 1 (10 points)

- Evaluate the creation of the dataset with at least 10,000 instances, assessing the presence and correct implementation of features typical for an e-commerce scenario. Check for the correct range and type (e.g., numeric, categorical) for each feature like Age, Gender, Income, etc.
- Assess how realistically the relationships between features are modeled (e.g., income correlating with average spending). The guidelines should check for logical assumptions and their implementation in the data generation process.
- Award points for effective visualizations that clearly demonstrate the relationships between features. Specifically look for the creation and visualization of non-linear relationships between features, which are crucial for testing the models in Part 2.

Guidelines 2 (15 points)

- Check for correct implementation and training of three different tree-based models: Decision Tree, Random Forest, and a third model of the student's choice (e.g., XGBoost, Bagging Classifier). Evaluate whether the basic parameters for each model are appropriately configured and documented.
- Grade based on the explanation provided for choosing each model, focusing on the appropriateness of each model for the dataset. This includes discussing the strengths of each model in handling complex data structures like those introduced in Part 1.
- Evaluate the choice and application of performance metrics (e.g., accuracy, precision, recall, F1 score, ROC-AUC). Look for a clear justification for each metric chosen and a thorough analysis of the models' performance on these metrics. Analysis of Non-Linear Performance: Grade based on the commentary on each model's ability to capture non-linear relationships. This should include a detailed discussion linking model performance back to the synthetic data features and the non-linear relationships that were visualized in Part 1.

Assignment 3: Predicting Sales Performance Using a Neural Network

This assignment is based on the `Sales_Win_Loss.csv` dataset attached to the exam. The dataset is from IBM, and is also available on Kaggle. Each record in the dataset represents an opportunity for a sales team and includes various attributes related to the sales process and the final outcome (win or loss).

The dataset contains attributes such as the type of client, size of the client, deal size, region, and whether the sales opportunity was won or lost. These features can be leveraged to understand patterns and predict future sales outcomes.

Exercise 3.1: Load the *Sales_Win_Loss* dataset. Familiarize yourself with its structure and the variables it includes. Conduct necessary data cleaning and preprocessing steps, such as dealing with missing values, encoding categorical features, and scaling numerical values. Divide the dataset into a training set and a testing set.

Exercise 3.2: Design a neural network with one hidden layer for predicting the 'win' or 'loss' outcome of sales opportunities. Describe the network architecture, including your choice of activation functions and the number of neurons in the hidden layer. Explain why these choices are appropriate.

Exercise 3.3: Train your neural network on the training data. Outline the training process, specifying the optimizer used, number of epochs, batch size, and performance metrics monitored during training. Explain why the chosen metrics are suitable for assessing the performance of a model in this application.

Exercise 3.4: Evaluate your model's performance on the test set using appropriate metrics (e.g., accuracy, precision, recall, F1-score). Present and interpret the evaluation results, highlighting the neural network's ability to predict sales outcomes effectively.

Exercise 3.5: Reflect on the performance of your neural network. Discuss potential improvements or alternative modeling techniques that could enhance prediction accuracy. Consider whether a neural network is the optimal model for this type of prediction problem. If not, suggest other models that might be more effective, providing reasons for your choices.

Guidelines

This assignment counts for 25% of the total grade.

Guideline 3.1 (5 points)

- Check for correct loading of the Sales_Win_Loss.csv dataset and an initial analysis of its structure. Evaluate the completeness of the description of the dataset's variables and their types.
- Assess the implementation of necessary data preprocessing steps, such as handling missing values, encoding categorical features, and scaling numerical values. Award points based on the appropriateness of the methods used and their justification.
- Grade the method used to divide the dataset into training and testing sets, ensuring the split is rational and justified, with appropriate proportions.

Guidelines 3.2 (5 points)

- Evaluate the design of the neural network, particularly the choice of the number of neurons in the hidden layer and the activation functions used. Points should be given based on the appropriateness and explanation of these choices.
- Look for clear reasoning behind each design choice, including how these choices are expected to influence the model's performance on this specific problem.
- Award points for how clearly and deeply the student explains the neural network architecture, showing an understanding of how the architecture impacts learning and performance.

Guideline 3.3 (5 points)

- Check for a clear outline of the neural network training process, including the optimizer used, number of epochs, batch size, and the rationale behind these choices. Performance Metrics: Evaluate the choice of performance metrics monitored during training. Points should be given for selecting suitable metrics and providing a rationale that aligns with the goals of a sales prediction model. Explanation and Justification: Grade the depth of explanation regarding how the training process and metrics are optimized to achieve effective learning and prediction accuracy.

Guideline 3.4 (5 points)

- Assess the application and explanation of metrics such as accuracy, precision, recall, and F1-score for evaluating the model on the test set.
- Look for a clear presentation and thoughtful interpretation of the evaluation results. Points should be awarded for insights into how well the neural network predicts sales outcomes, highlighting strengths and potential weaknesses.
- Grade the depth of analysis in discussing how the model's performance metrics relate to its ability to predict real-world sales scenarios.

Guidelines 3.5 (5 points)

- Evaluate the reflection on the neural network's performance, including potential reasons for any limitations observed.
- Look for thoughtful suggestions on how to improve the model's performance, including consideration of alternative modeling techniques.
- Grade the discussion on whether a neural network is the optimal choice for this prediction problem. Points should be given for well-reasoned arguments for or against the use of neural networks, including suggestions of other models that might be more effective and why.

Assignment 4: Reflection on Your Mini Project

For each of the following 5 questions, write a response of no more than 300 words.

Exercise 4.1: Briefly describe the problem or research question you tackled in your mini project. Why did you choose this particular problem?

Exercise 4.2: Discuss the process of data collection and preprocessing. Reflect on how the quality and nature of data influenced your project results.

Exercise 4.3: Which machine learning algorithm(s) did you choose for your project, and why? Describe the process of implementing the algorithm and any difficulties you encountered.

Exercise 4.4: Were you able to effectively address the problem identified? What insights did you gain from the analysis?

Exercise 4.5: If you were to continue this project, what aspects would you focus on improving? Discuss any additional experiments or changes in methodology you would consider for improving your project.

Guidelines

This assignment counts for 25% of the total grade.

- The responses should provide details of the mini project presented in class.

Guidelines 4.1 (5 points)

- Evaluate the clarity with which the student describes the problem or research question tackled in their mini project. Award points for a concise yet comprehensive explanation of the project's scope.
- Grade based on the depth of reasoning provided for choosing the specific problem or research question. Look for personal or academic motivations, relevance to current issues, or potential impact.

Guidelines 4.2 (5 points)

- Assess how well the student describes the data collection and preprocessing steps. Points should be awarded for detailed explanations of the methods used and the rationale behind them.
- Grade the reflection on how the quality and nature of the data influenced the results of the project. Look for insights into challenges faced due to data quality and how they were addressed.
- Evaluate the depth of analysis regarding the impact of data on the project's outcomes. Reward responses that show a clear understanding of the connection between data quality and project success.

Guidelines 4.3 (5 points)

- Check for a clear explanation of the machine learning algorithm(s) chosen for the project. Points should be awarded for justifying why specific algorithms were suitable for addressing the project's problem.
- Assess the description of the algorithm implementation process, including any difficulties encountered. Grade the thoroughness and honesty in discussing challenges and how they were overcome.
- Reward depth of understanding and ability to communicate complex machine learning concepts clearly.

Guidelines 4.4 (5 points)

- Evaluate how well the student articulates whether and how they were able to address the identified problem. Points should be based on the clarity and specificity of the outcomes discussed.
- Grade the quality and depth of the insights gained from the analysis. Look for detailed examples of findings and their implications on the problem studied.
- Check for analytical depth in linking project results to broader concepts or potential real-world applications.

Guidelines 4.5 (5 points)

- Assess the student's ability to critically identify areas of their project that could be improved. Points should be awarded for specific, actionable suggestions.
- Evaluate the feasibility and creativity of proposed additional experiments or changes in methodology. Reward clear, innovative ideas that could realistically enhance the project.
- Grade the overall reflective quality of the response, rewarding responses that demonstrate a thoughtful consideration of the project's future potential and areas for growth.